CYCLISTIC BIKE-SHARE ANALYSIS

Member vs Casual Rider Behavior — 2023

Capstone Project Report

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Dataset period: January 2023 – December 2023
Final cleaned records used: 5,714,805

Executive Summary

This project analyzes 2023 bike-share trip data to understand how casual riders and annual members use the bike-sharing service differently. The main business objective is to identify behavioral patterns that can help the company develop marketing strategies for converting casual riders into annual members.

The analysis used 12 monthly CSV files and a final cleaned dataset of 5,714,805 records. Data preparation included validation of row counts, duplicate investigation, missing-value assessment, date/time standardization, conversion of date columns to DATETIME, and removal of rides with zero or negative duration. Long rides exceeding 24 hours were treated as outliers and excluded from the analytical view.

The results show a clear behavioral difference between the two groups. Members generated 3,655,725 rides, while casual riders generated 2,052,662. However, casual riders had a much longer average ride duration (20.70 minutes) than members (12.07 minutes). Members showed stronger weekday and commuting-hour activity, while casual riders were more active on weekends and around leisure-oriented stations. Both groups showed strong seasonal growth during spring and summer.

Dataset Source: The dataset used in this project is the Cyclistic Bike-Share dataset provided through the Google Data Analytics Professional Certificate Capstone Case Study.

Original Dataset:

https://divvy-tripdata.s3.amazonaws.com/index.html

Key Findings at a Glance

Metric	Member	Casual	Main Interpretation
Total rides	3,655,725	2,052,662	Members account for more rides overall.
Average ride length	12.07 min	20.70 min	Casual riders take substantially longer rides.
Peak day	Thursday: 588,688	Saturday: 409,374	Members are more weekday-oriented; casual riders are more weekend-oriented.
Peak month	August: 460,240	July: 330,334	Both groups are strongest in the warmer months.
Peak hour	5 PM: 387,462	5 PM: 198,755	Evening demand is high for both groups; members also show a strong morning peak.

1. Introduction

Bike-sharing systems generate large volumes of trip data that can be used to understand customer behavior, service demand, and opportunities for customer retention. This project focuses on the difference between two important customer groups: casual riders and annual members.

Casual riders generally use the service without an annual membership, whereas members have made a longer-term commitment to the service. Understanding where, when, and how these groups ride can help the company design targeted campaigns instead of using the same marketing approach for every customer.

2. Business Task

The primary business task is to identify how casual riders and annual members use the bike-sharing service differently and use those differences to recommend strategies that may encourage casual riders to purchase annual memberships.

The analysis focuses on the following questions:

- How many rides are taken by members and casual riders?
- How does average ride duration differ between the two groups?
- Which bike types are preferred by each membership group?
- Which days and months have the highest ride activity?
- At which hours are members and casual riders most active?
- Which start and end stations are most frequently used by each group?
- What marketing opportunities can be identified from these behavioral patterns?

3. Data Description

The project uses 12 monthly CSV files covering rides during 2023. The data was imported into MySQL for validation, cleaning, transformation, and SQL-based analysis. The main fields include ride ID, bike type, start and end timestamps, start and end station information, latitude/longitude fields, and membership type.

Field	Purpose
ride_id	Unique identifier used to identify individual rides.
rideable_type	Identifies the type of bike used.
started_at / ended_at	Used to calculate ride duration and analyze time patterns.
start_station_name / end_station_name	Used for station-level analysis.
start_lat / start_lng / end_lat / end_lng	Geographic coordinates associated with stations.
member_casual	Identifies the rider as a member or casual rider.

Dataset Size

The validation workbook records 5,719,877 imported rows. After data cleaning, including removal of 5,072 rides with zero or negative duration, the final dataset contained 5,714,805 records.

4. Data Cleaning and Validation

Data cleaning was performed before the business analysis so that calculations were based on consistent date/time values and valid ride durations. The following validation steps were documented.

Step	Validation / Cleaning Activity	Result	Action
1	Total rows	5,719,877 rows imported	All 12 monthly files were imported.

2–3	Ride ID uniqueness / duplicate investigation	Duplicate difference detected; five sample ride IDs appeared twice	Duplicates were investigated before deletion.
4	Missing values	875,716 missing start-station names; 929,202 missing end-station names; coordinates and member type complete	Station-name blanks were retained for investigation rather than removed automatically.
5	Date range	2023-01-01 through 2024-01-01 end timestamps	Rides beginning on Dec. 31, 2023 may end on Jan. 1, 2024.
6–7	Date standardization	190,301 January records converted; 0 old-format rows remained	Completed.
8–9	Invalid ride duration	5,072 zero/negative-duration rides identified and deleted	Completed.
10	Final row count	5,714,805 records	Used as the cleaned dataset.
11	Datetime data type	started_at and ended_at converted to DATETIME	Completed.

Missing Values

Missing values were concentrated in station-name fields. Start-station names had 875,716 missing values and end-station names had 929,202 missing values. Latitude/longitude fields and membership type were reported as complete. Because missing station names do not necessarily mean that a ride record is unusable for membership, duration, or time analysis, these records were retained rather than deleting large portions of the dataset.

Ride Duration and Outlier Investigation

Initial ride-duration analysis showed a minimum of 0.02 minutes, a maximum of 98,489.07 minutes, and an overall average of 18.20 minutes. A total of 6,418 rides were longer than 24 hours, representing approximately 0.11% of the cleaned records. These unusually long rides were treated as outliers and excluded from the analytical view so that extreme values would not distort normal ride behavior.

5. Analysis Methodology

SQL was used to calculate ride counts, average durations, time-based patterns, bike-type usage, and station rankings. The cleaned data was then exported into analysis files and visualized in Tableau. Two Tableau dashboards were created: Rider Behavior Analysis and Station Analysis.

- Membership comparison: count rides by member/casual status.
- Ride-duration comparison: calculate average ride length in minutes.
- Bike-type comparison: count rides by bike type and membership.
- Day-of-week analysis: compare ride volume and average duration across Sunday–Saturday.
- Monthly analysis: compare ride volume across all 12 months.
- Hourly analysis: compare ride volume across start hours 0–23.
- Station analysis: rank the top start and end stations separately for casual riders and members.
- Visualization: present the findings using bar charts and line charts in Tableau.

6. Analysis Results

6.1 Ride Count by Membership

Members generated 3,655,725 rides, compared with 2,052,662 rides by casual riders. This shows that annual members produced the larger share of total ride activity in the 2023 dataset.

6.2 Average Ride Length

The average ride length for members was 12.07 minutes, while the average for casual riders was 20.70 minutes. Therefore, casual rides were approximately 8.63 minutes longer on average. This difference is important because ride frequency and ride duration describe different types of customer behavior: members ride more frequently, while casual users tend to spend longer on each trip.

6.3 Bike Type Preference

Membership	Bike Type	Ride Count
Casual	Electric bike	1,102,928
Casual	Classic bike	872,914
Casual	Docked bike	76,820
Member	Electric bike	1,838,568
Member	Classic bike	1,817,157

Electric bikes were the most-used bike type for both groups. Casual riders also used docked bikes, while the recorded member results contained electric and classic bike usage but no docked-bike usage.

6.4 Day-of-Week Behavior

Group	Highest Day	Rides	Lowest Day	Rides
Member	Thursday	588,688	Sunday	408,285
Casual	Saturday	409,374	Monday	234,099

Members show stronger activity during weekdays, while casual riders show stronger activity on weekends. This pattern is consistent with a likely difference between routine transportation/commuting and recreational or leisure riding.

6.5 Monthly Ride Trends

Group	Peak Month	Peak Rides	Lowest Month	Rides
Member	August	460,240	January	147,151
Casual	July	330,334	January	39,221

Both groups experienced strong growth from spring into summer and a decline during the later months of the year. This indicates a strong seasonal component in bike-share demand.

6.6 Hourly Ride Trends

The highest recorded hour for both groups was 5 PM: 387,462 member rides and 198,755 casual rides. Members also show a pronounced morning peak, which supports the interpretation of stronger commuting-related usage. Casual activity is more concentrated from afternoon into evening, which is more consistent with leisure-oriented usage.

6.7 Average Ride Length by Day

Day	Casual (min)	Member (min)
Sunday	24.16	13.45
Monday	20.34	11.46
Tuesday	18.52	11.60
Wednesday	17.66	11.52
Thursday	18.06	11.59
Friday	20.10	12.00
Saturday	23.45	13.45

Casual riders had their longest average rides on Sunday and Saturday, while members maintained relatively shorter and more stable ride durations. This reinforces the difference between recreational riding and routine transportation behavior.

7. Station Analysis

Station-level analysis was performed separately for casual riders and members to identify locations with the highest ride activity. The station results show one of the clearest behavioral differences between the two groups.

7.1 Top Start Stations — Casual Riders

Rank	Station	Rides
1	Streeter Dr & Grand Ave	45,817
2	DuSable Lake Shore Dr & Monroe St	30,347
3	Michigan Ave & Oak St	22,594
4	DuSable Lake Shore Dr & North Blvd	20,289
5	Millennium Park	20,091
6	Shedd Aquarium	17,721
7	Theater on the Lake	16,317
8	Dusable Harbor	15,412
9	Wells St & Concord Ln	12,147
10	Montrose Harbor	11,959

Casual riders frequently began trips at lakefront, park, and attraction-oriented locations. This pattern strongly suggests recreational and sightseeing use.

7.2 Top Start Stations — Member Riders

Rank	Station	Rides
1	Clinton St & Washington Blvd	26,165
2	Kingsbury St & Kinzie St	26,136
3	Clark St & Elm St	24,970
4	Wells St & Concord Ln	21,400
5	Clinton St & Madison St	20,568
6	Wells St & Elm St	20,368
7	University Ave & 57th St	20,002
8	Broadway & Barry Ave	18,943
9	Loomis St & Lexington St	18,877
10	State St & Chicago Ave	18,428

Member start stations are more concentrated around business, residential, and transit-connected areas, supporting the interpretation that members use the service more routinely for daily transportation.

7.3 Top End Stations

Group	Highest End Station	Rides
Casual	Streeter Dr & Grand Ave	49,304
Casual	DuSable Lake Shore Dr & Monroe St	27,536
Member	Clinton St & Washington Blvd	27,402
Member	Kingsbury St & Kinzie St	26,344

The end-station results reinforce the same pattern observed in start stations: casual riders frequently finish around recreational and lakefront locations, while members frequently finish around commuter-oriented locations.

8. Key Findings and Business Insights

Members generate more total rides. The member group recorded 3.66 million rides versus 2.05 million casual rides, showing a much larger volume of activity from the existing member base.

Casual riders take longer trips. Average casual ride duration was 20.70 minutes versus 12.07 minutes for members. The longer duration is especially strong on weekends.

Members show a weekday/commuting pattern. Member ride counts are highest on Thursday and the hourly pattern shows strong morning and evening activity.

Casual riders show a weekend/leisure pattern. Saturday is the strongest day for casual riders, and casual start/end stations are concentrated around attractions, parks, and waterfront areas.

Demand is seasonal. Ride activity rises sharply in spring and summer and falls toward winter, so membership campaigns can be timed around the period when casual usage is increasing.

Electric bikes are highly relevant to both groups. Electric bikes were the most-used bike type for both casual riders and members, making them a useful feature to emphasize in membership messaging.

9. Business Recommendations

1. Target frequent casual riders with membership offers. Casual riders already generated more than two million trips in the dataset. Marketing should focus on converting the most active casual users rather than treating every casual rider equally. Personalized offers can be based on ride frequency and recent usage.

2. Promote annual membership before and during the spring/summer growth period. Casual ride volume rises strongly from March through July. Membership promotions should become more visible before this high-demand period so that users are encouraged to convert while their engagement is increasing.

3. Use weekend-focused membership messaging. Casual riders are strongest on Saturday and have their longest average rides on Sunday and Saturday. Campaigns can emphasize how an annual membership can reduce repeated trip costs and provide convenient access for regular weekend riders.

4. Use commuter-oriented messaging for users showing weekday patterns. Some casual riders may already be using the service at times and locations associated with regular transportation. These users are strong membership prospects and can be targeted with convenience, savings, and unlimited-access messaging.

5. Highlight electric-bike availability. Electric bikes are the most popular bike type for both groups. Membership campaigns can emphasize access to popular electric-bike options and the convenience of using them for longer or repeated trips.

6. Use location-based campaigns near popular casual stations. Casual activity is concentrated around locations such as Streeter Dr & Grand Ave, DuSable Lake Shore Dr, Millennium Park, and other lakefront or attraction areas. Digital promotions, QR codes, or station-area advertising near these locations could reach casual users at the point of use.

7. Test different offers and measure conversion. Rather than using one permanent offer, Cyclistic should test different membership messages and incentives, then compare conversion rates. This would allow the company to identify which strategy produces the strongest return.

10. Tableau Dashboard

The final analysis was visualized in Tableau using two dashboards. The dashboards were designed to make the differences between members and casual riders easy to compare.

Dashboard	Included Visualizations
Rider Behavior Analysis	Total rides by membership, average ride length, monthly ride trends, ride count by day, hourly ride trends, bike type usage, and average ride length by day.
Station Analysis	Top start stations for casual riders, top start stations for members, top end stations for casual riders, and top end stations for members.

The dashboards use consistent member/casual color coding and provide a visual summary of the analysis. The first dashboard focuses on behavior over time and ride characteristics, while the second focuses on location.

Interactive Tableau Dashboard:
https://public.tableau.com/app/profile/husnain.afzal/viz/CyclisticBikeShareAnalysis2023

11. Limitations

- The analysis is based on one calendar year (2023), so longer-term trends cannot be confirmed from this dataset alone.
- A large number of station-name fields are missing, which limits some station-level analysis.
- Very long rides above 24 hours were treated as outliers and excluded from the analytical view.
- The analysis describes behavioral associations; it does not prove that a particular factor directly causes a rider to become or remain a member.
- The available trip data does not by itself explain customer motivations, income, demographics, or reasons for choosing membership.

12. Conclusion

The 2023 analysis shows a clear difference between annual members and casual riders. Members generated substantially more rides and showed stronger weekday and commuting-hour activity, while casual riders took longer trips, were more active on weekends, and were strongly associated with recreational and lakefront locations.

These differences provide a practical basis for membership marketing. The strongest opportunity is not simply to advertise annual memberships to all casual riders, but to identify high-engagement casual users and target them with messages that match their behavior. Seasonal campaigns, weekend-focused offers, commuter-oriented messaging, electric-bike promotion, and location-based marketing near popular casual stations can all be tested as conversion strategies.

Overall, the project demonstrates how data cleaning, SQL analysis, Excel-based validation, and Tableau visualization can be combined to turn millions of individual trip records into actionable business insights.

Appendix A — Main Analytical Results

Analysis	Key Result
Final cleaned dataset	5,714,805 records
Member rides	3,655,725
Casual rides	2,052,662
Member average ride	12.07 minutes
Casual average ride	20.70 minutes
Long rides >24 hours	6,418 (0.11%)
Invalid durations removed	5,072
Member peak month	August — 460,240 rides
Casual peak month	July — 330,334 rides
Member peak hour	5 PM — 387,462 rides
Casual peak hour	5 PM — 198,755 rides

Appendix B — Deliverables Used

- Data_Validation_Results.xlsx — data quality and cleaning checks.
- Analysis_Results.xlsx — ride-duration and outlier analysis.
- Business_Analysis.xlsx — membership, time, bike, station, and behavioral analysis.
- Tableau Rider Behavior Analysis dashboard.
- Tableau Station Analysis dashboard.